

Elevator Preliminary Requirements

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Here is a first stab at the primary requirements for our new elevator application.

INSIDE THE CABIN

The following requirements are described from the perspective of a user inside of a cabin.

Capabilities What can a user do while inside of a cabin?

Go to floor (number) - this is probably achieved by pressing a button corresponding to the desired destination.

Open/Close door - probably a button

Manually block door - if user obstructs door while not fully closed, the door will quickly (within n milliseconds) reopen fully for n seconds.

Feedback The following information will be available to the user inside the Cabin:

Current location (nearest floor) - a number displayed independently or on some sort of scale.

Current service direction - probably a lighted arrow of some sort.

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Service status - whether or not this cabin is available for service or in maintenance mode.

Perceived performance Cabin moves as fast as possible, but never exceeds $\pm N$ g's. (Exact value to be determined later). The cabin should always move toward the passenger's desired floor (does not double back assuming the passenger entered the cabin going the right way in the first place).

Door never stays open longer than n seconds. Door never closes faster than n seconds. Exact values to be determined later.

AT A FLOOR

Capabilities This is what the passenger can do while outside of the cabin:

Call Up/Down - the user calls the elevator with the intention of traveling up or down. At the top floor only the down call is available and at the bottom floor only the up call is available.

Feedback The user has the following information available at a floor:

Current location of each cabin

Current service direction of each cabin (this information need only be available when the cabin is about to make a stop at the current floor)
The service direction must be available within n seconds of the impending stop.

Perceived performance Passenger should never have to wait more than n seconds after calling a cabin. Average time should be n seconds. There is a function where wait times is dependent on the following factors: number of floors serviced per shaft, number of shafts per bank in service.

Inside the cabin, user can select destination floor. Cannot cancel a floor destination. Can emergency stop the elevator. Can unstop the elevator.

ADMINISTRATIVE

For a given set of shafts, any number of floors may be skipped to provide express service.

Exercise 1

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Any cabin or group of cabins may be taken out of service temporarily for maintenance.

An out of service cabin may be directed remotely to any floor.